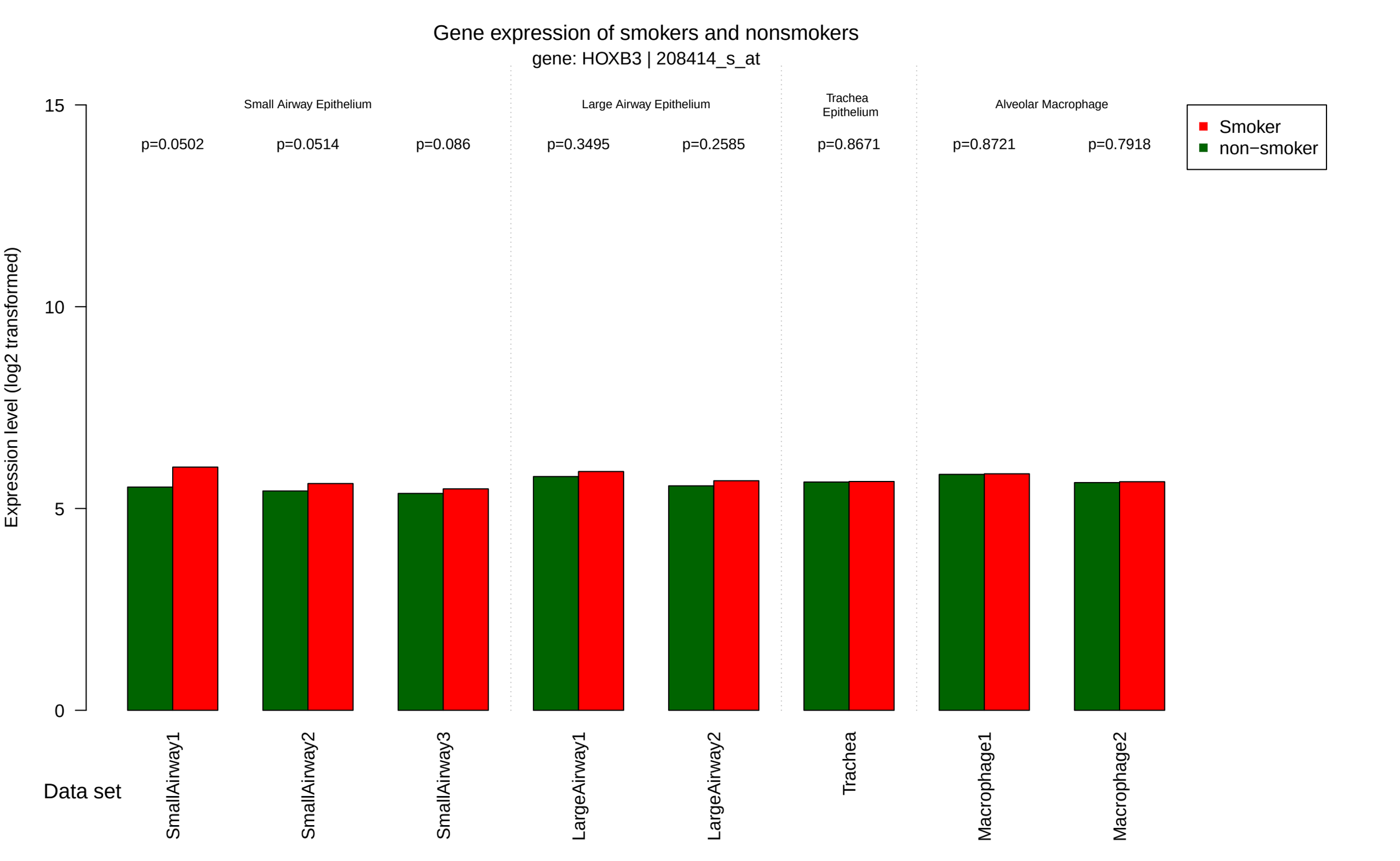
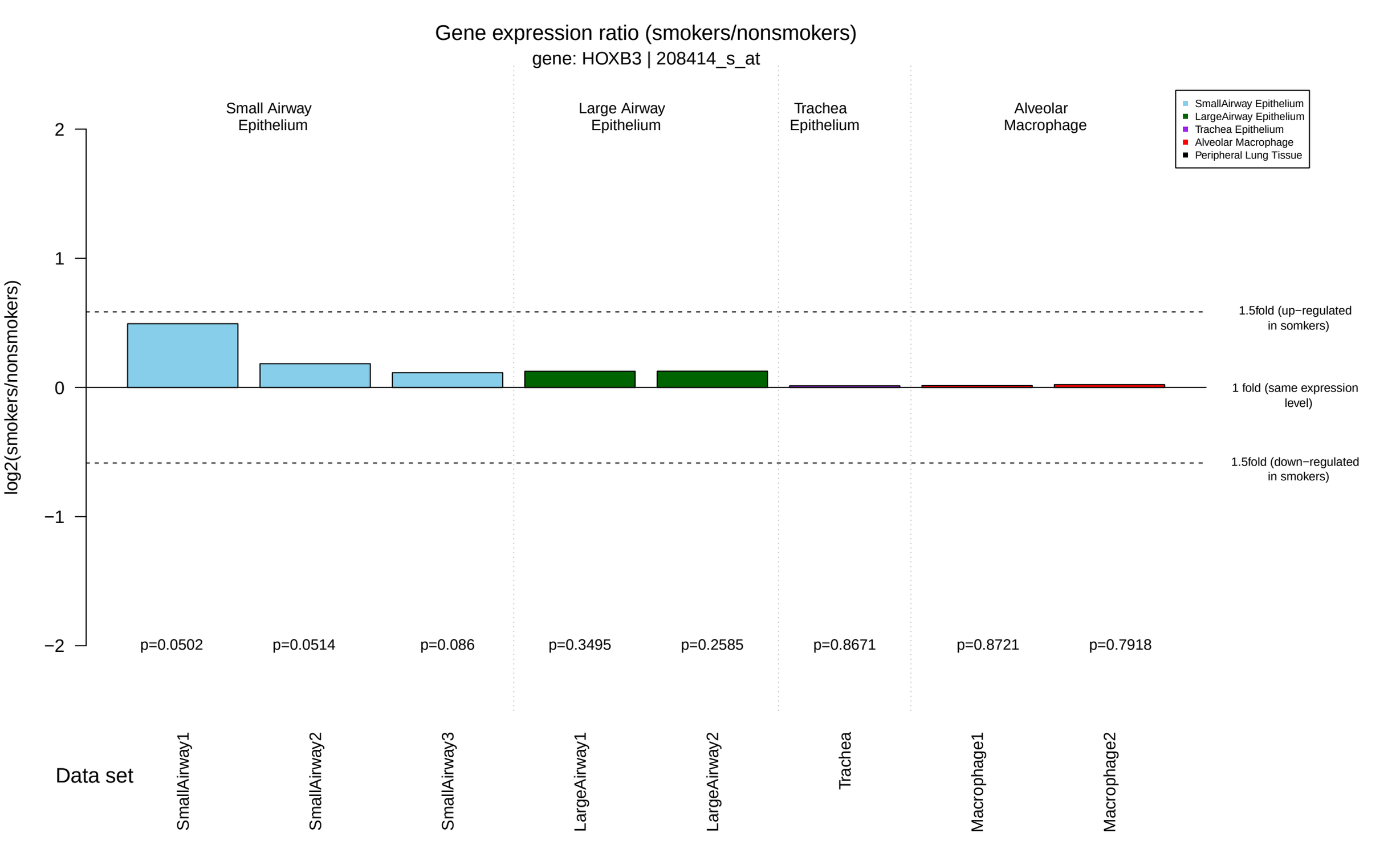
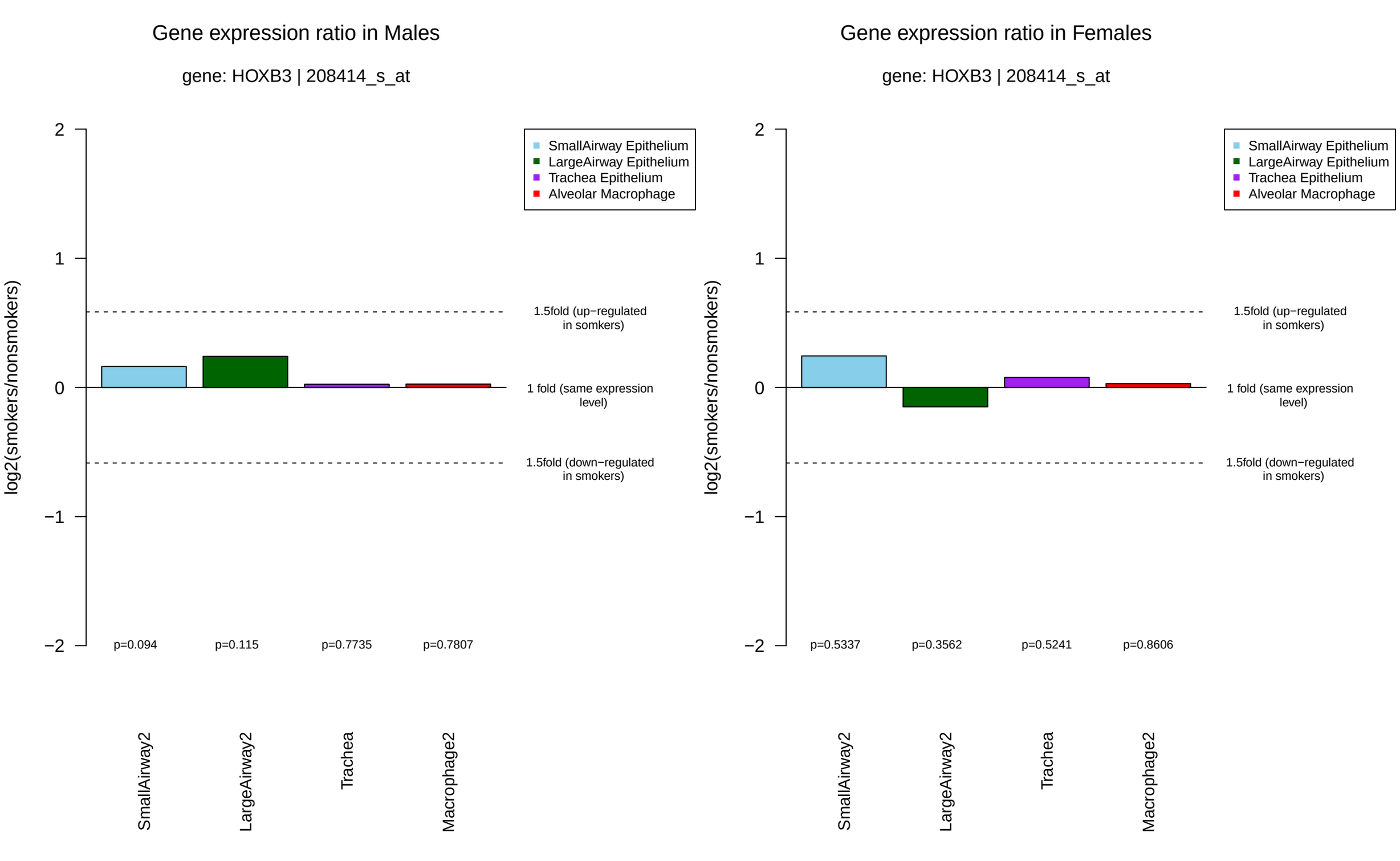
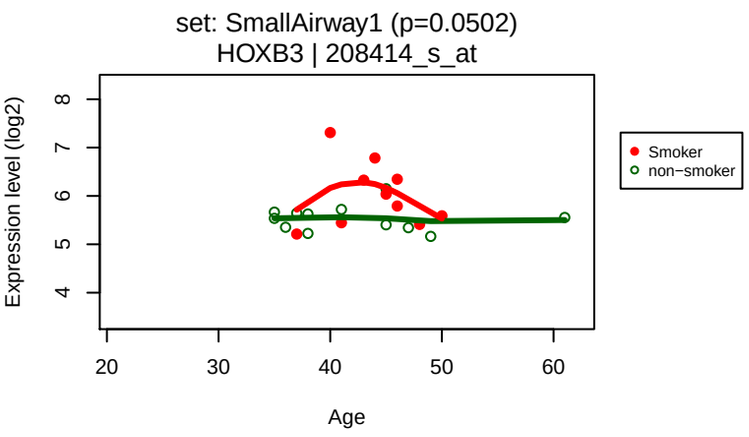
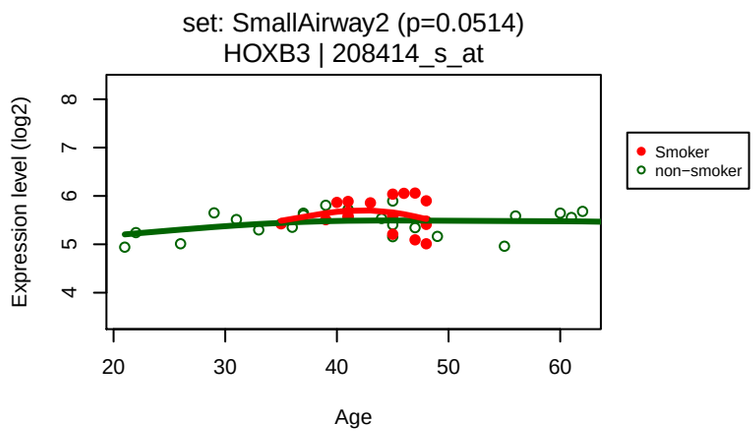
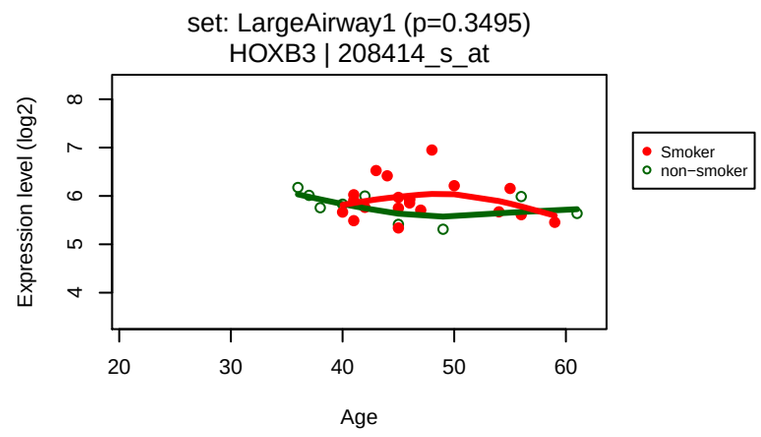

Gene expression of smokers and nonsmokers
gene: HOXB3 | 208414_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0502
p=0.0514
p=0.086
p=0.3495
p=0.2585
p=0.8671
p=0.8721
p=0.7918
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

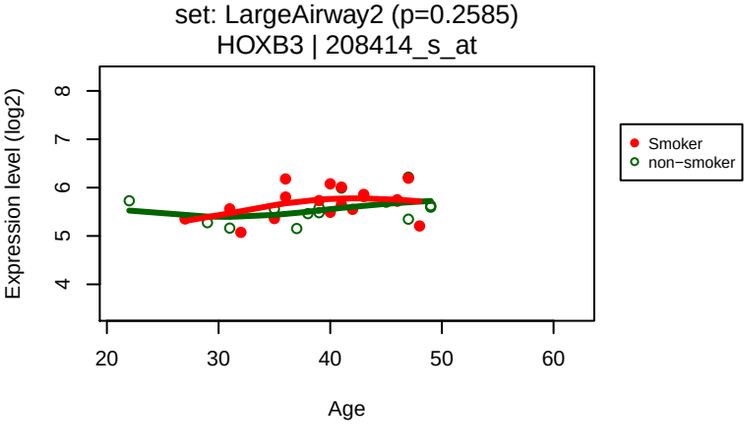

Gene expression ratio (smokers/nonsmokers)
gene: HOXB3 | 208414_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0502
p=0.0514
p=0.086
p=0.3495
p=0.2585
p=0.8671
p=0.8721
p=0.7918
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

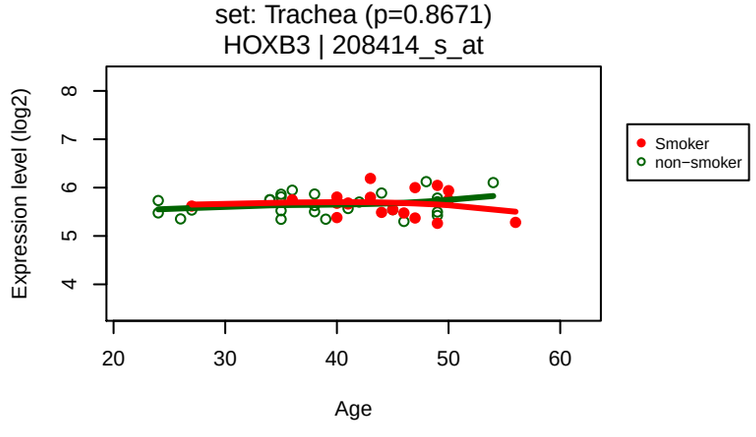

Gene expression ratio in Males
gene: HOXB3 | 208414_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.094
p=0.115
p=0.7735
p=0.7807
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXB3 | 208414_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5337
p=0.3562
p=0.5241
p=0.8606
SmallAirway2
LargeAirway2
Trachea
Macrophage2

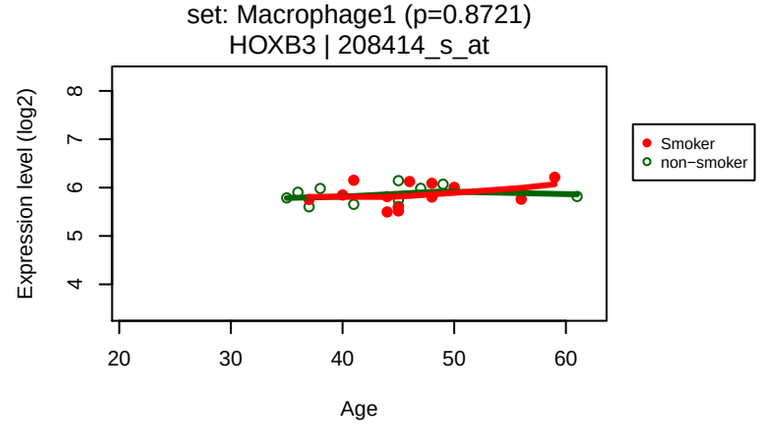

set: SmallAirway1 (p=0.0502)
HOXB3 | 208414_s_at
Expression level (log2)
Age
Smoker
non−smoker

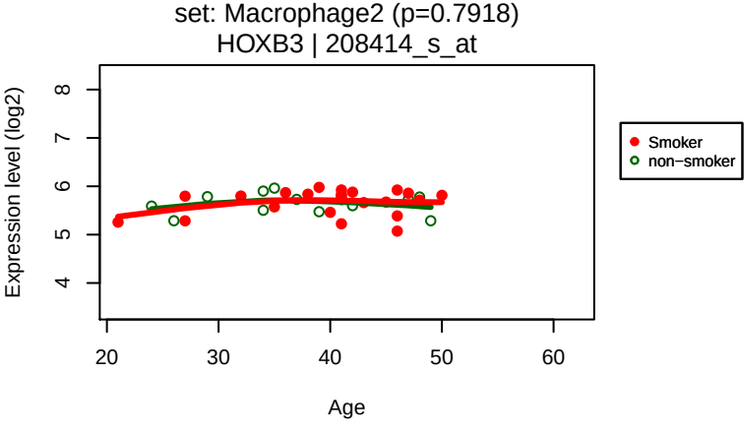

set: SmallAirway2 (p=0.0514)
HOXB3 | 208414_s_at
Expression level (log2)
Age
Smoker
non−smoker

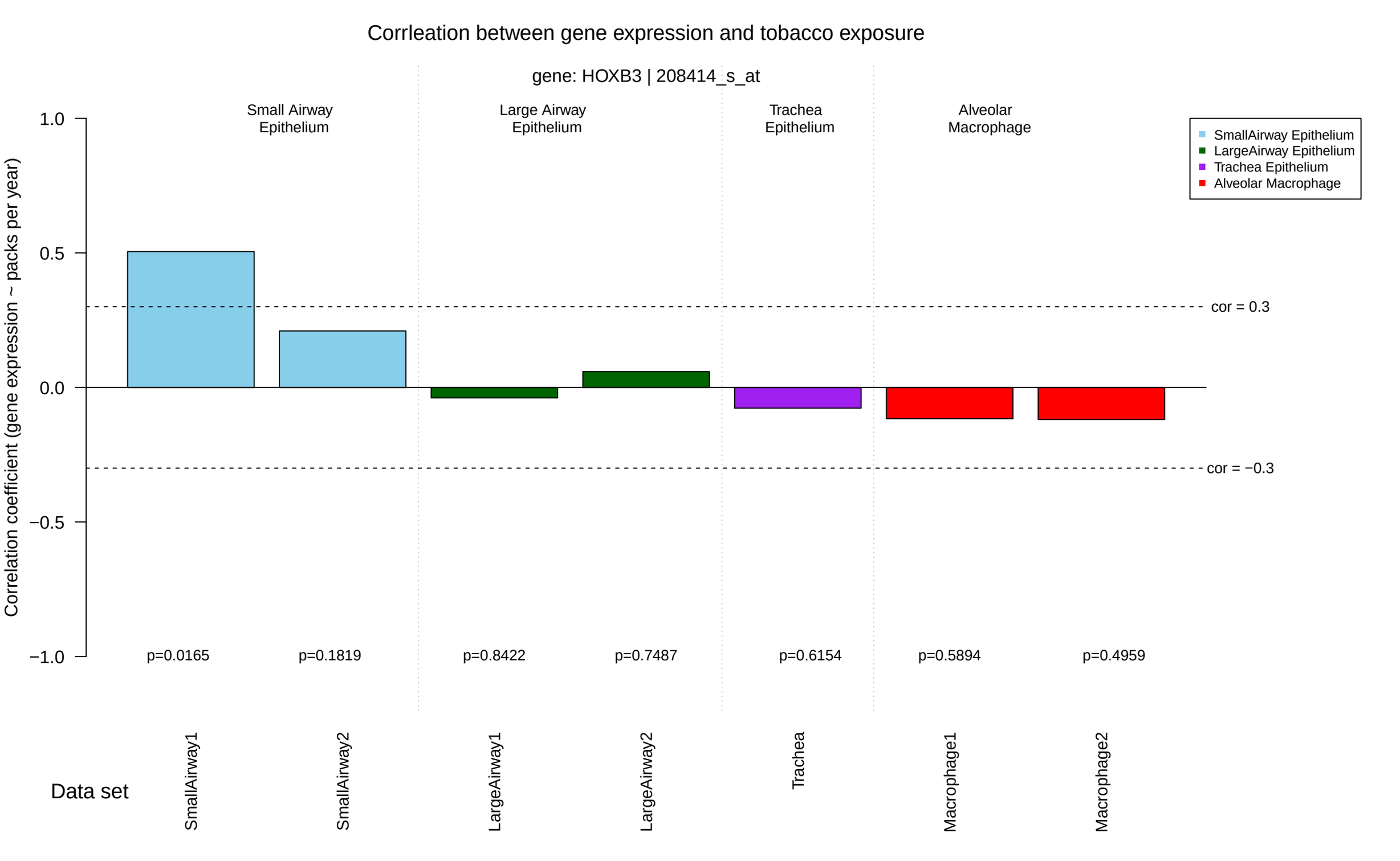

set: LargeAirway1 (p=0.3495)
HOXB3 | 208414_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2585)
HOXB3 | 208414_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8671)
HOXB3 | 208414_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8721)
HOXB3 | 208414_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7918)
HOXB3 | 208414_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXB3 | 208414_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0165
p=0.1819
p=0.8422
p=0.7487
p=0.6154
p=0.5894
p=0.4959
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2